20 Show that $\dfrac{(\sqrt{18}+\sqrt{2})^2}{\sqrt{8}-2}$ can be written in the form $a(b+\sqrt{2})$ where a and b are integers.

$\sqrt{18} = \sqrt{(9 \times 2)} = 3\sqrt{2}$

$(\sqrt{18} + \sqrt{2})^2 = (4\sqrt{2})^2 = 16 \times 2 = 32$

1 mark
Simplify the numerator exactly

$\dfrac{32}{\sqrt{8}-2} \times \dfrac{\sqrt{8}+2}{\sqrt{8}+2}$

1 mark
Multiply by the conjugate fraction

	$\sqrt{8}$	2
$\sqrt{8}$	8	$2\sqrt{8}$
-2	$-2\sqrt{8}$	-4

Denominator $= 8 + 2\sqrt{8} - 2\sqrt{8} - 4 = 4$

$\dfrac{32(\sqrt{8}+2)}{4} = 8(\sqrt{8}+2)$

$= 8(2\sqrt{2}+2) = 16(1+\sqrt{2})$

So $a = 16$ and $b = 1$.

1 mark
Exact required form reached

(Total for Question 20 is 3 marks)

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